Análise de Sistemas Elétricos II

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Centro de Engenharias e Ciências Exatas

Foz do Iguaçu

Universidade Estadual do Oeste do Paraná

FALTAS SIMÉTRICAS

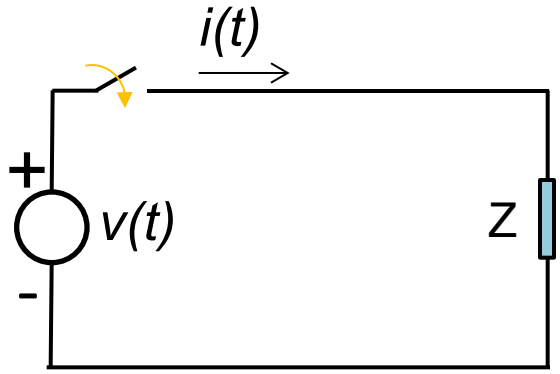
- Transitórios em Circuitos R-L
- Curto-Circuito Trifásico em Máquinas Síncronas sem Carga
- Máquinas Síncronas com Carga sob Condições Transitórias
- Faltas Trifásicas em Sistemas de Potência
- Uso da Matriz Impedância de Barra para Calcular as Correntes de Curto-Circuito

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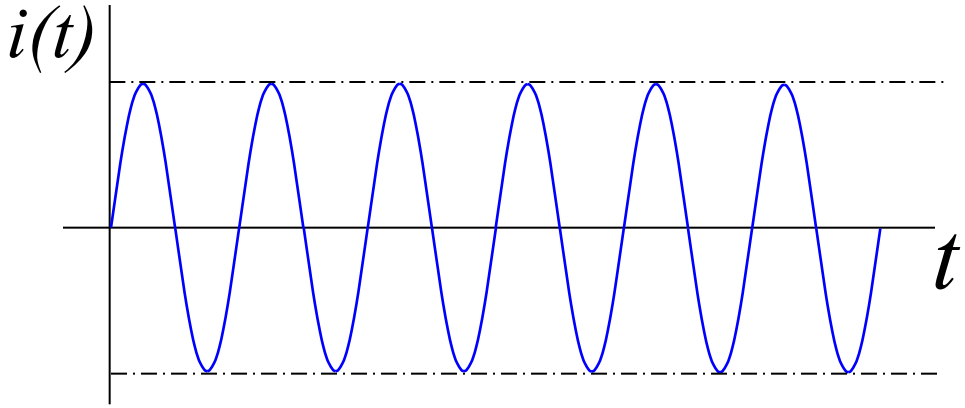
Transitórios em Circuitos R-L

Seja uma impedância constante e no instante $t = 0$ é aplicada uma tensão alternada:

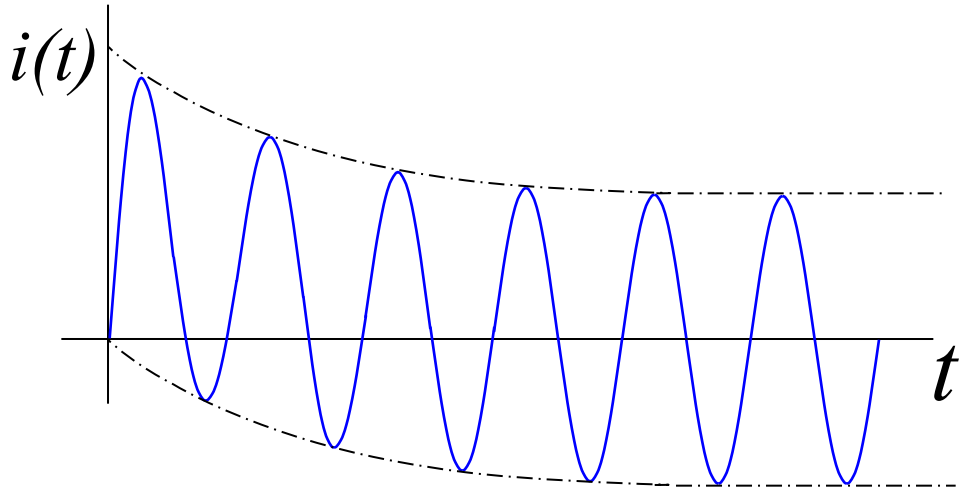
$$v(t) = V_{max}\sin(\omega t + \alpha) = Ri + L\frac{di}{dt}$$



$$i(t) = \frac{V_{max}}{|Z|}[\sin(\omega t + \alpha - \varphi) - e^{-Rt/L}\sin(\alpha - \varphi)]$$



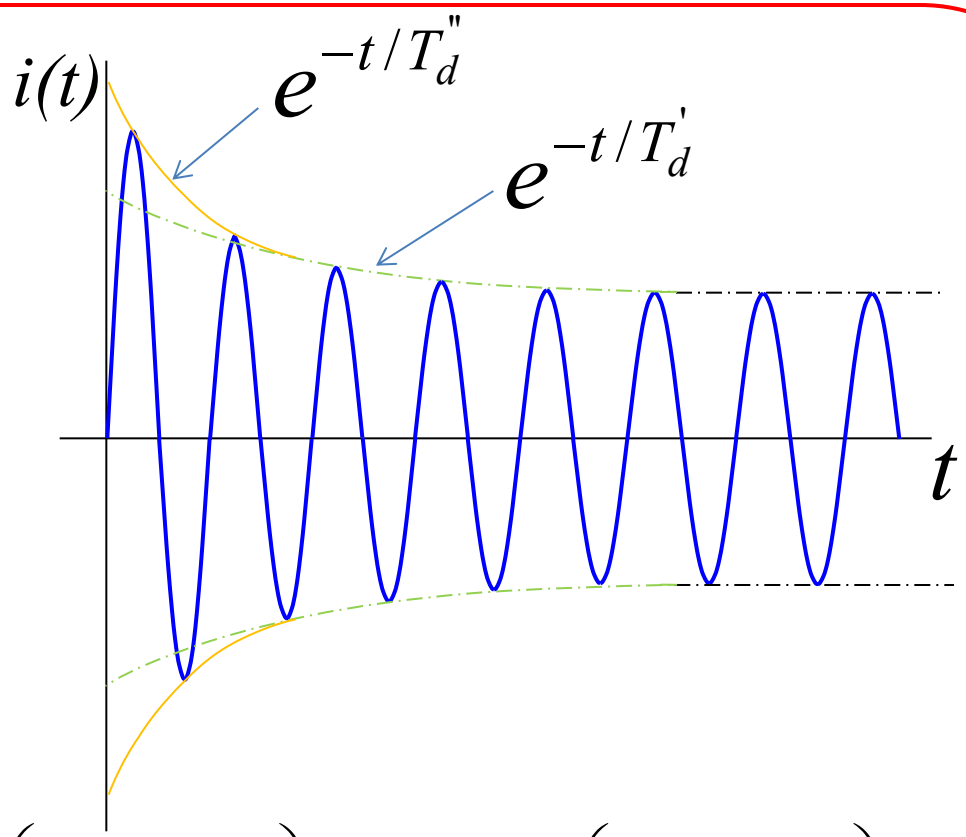
$\alpha - \varphi = 0$



$\alpha - \varphi = -\pi/2$

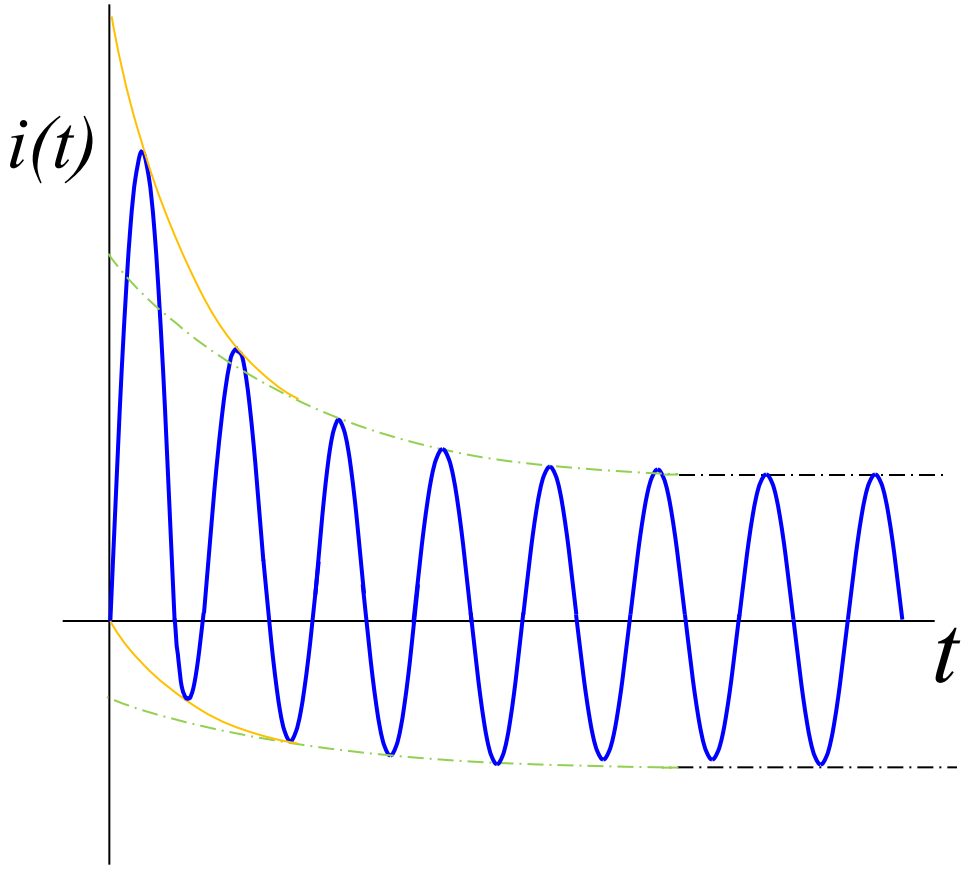
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Transitório em Geradores Síncronos



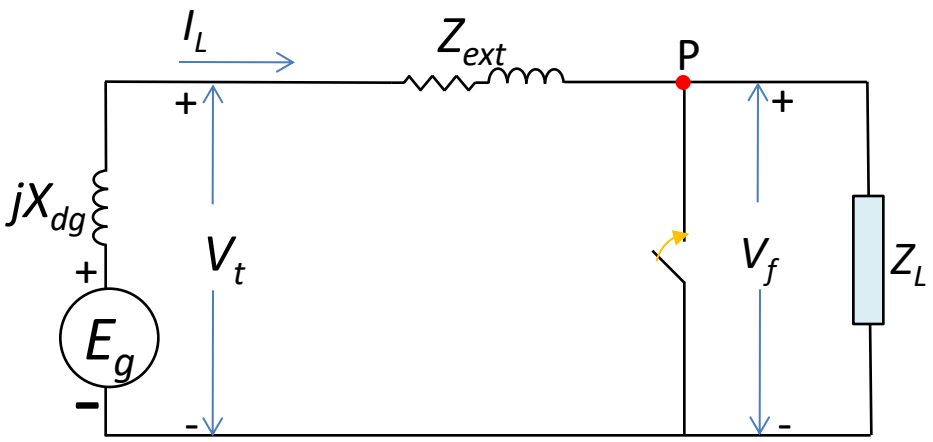
$$I(t) = |E_i|\frac{1}{X_d} + |E_i|\left(\frac{1}{X_d'} - \frac{1}{X_d}\right)e^{-t/T_d'} + |E_i|\left(\frac{1}{X_d''} - \frac{1}{X_d'}\right)e^{-t/T_d''}$$

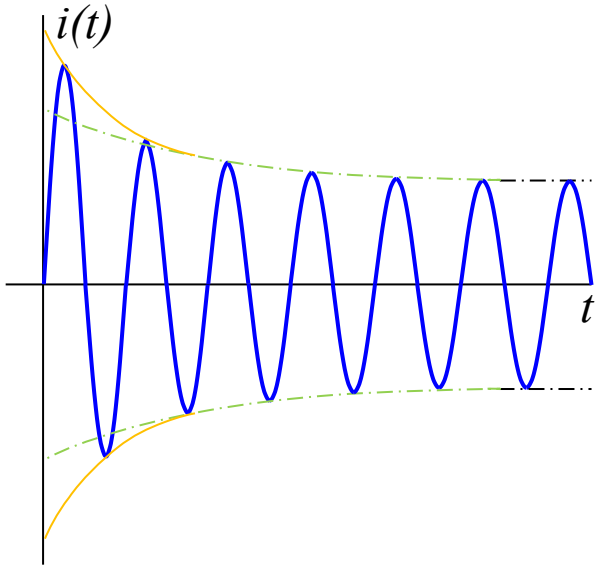
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Tensão interna do Gerador Síncrono sob condições de falta



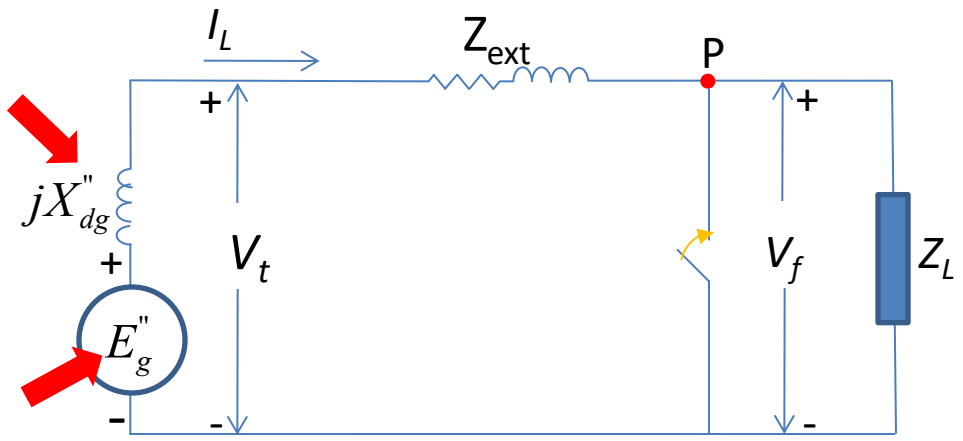


$X_d^{'}$: reatância transitória

$X_d^{''}$: reatância subtransitória

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Circuito para calcular *I"*



$$E''_g = V_t + jX''_{dg} I_L = V_f + (Z_{ext} + jX''_{dg}) I_L$$

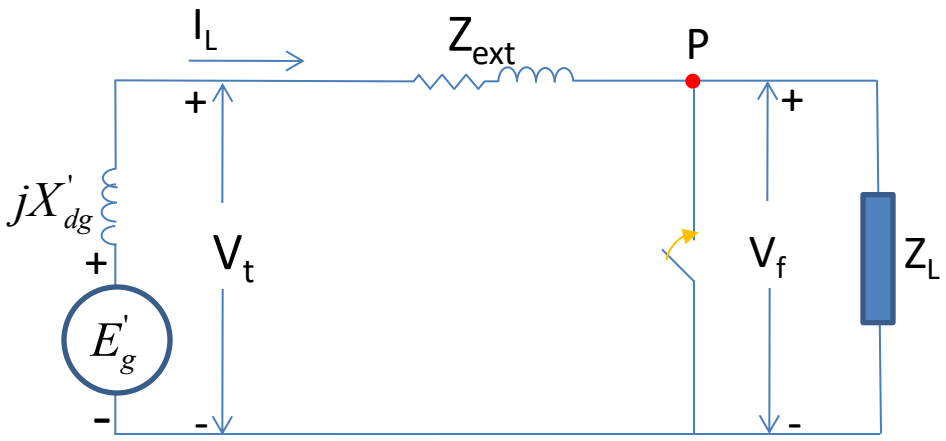
Regime permanente

$$I''_g = \frac{E''_g}{Z_{ext} + jX''_{dg}} = \frac{V_f}{Z_{ext} + jX''_{dg}} + I_L$$

Curto-circuito

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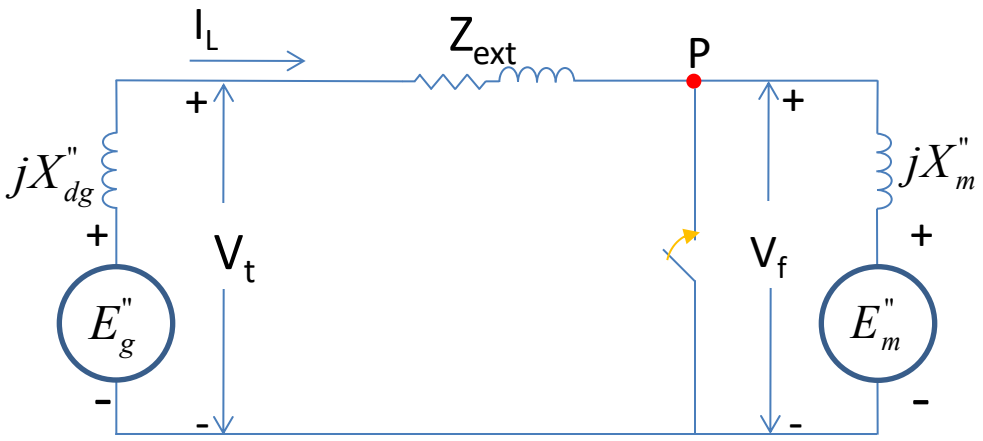
Circuito para calcular *I'*



$$E_g^{'} = V_t + jX_{dg}^{'} I_L = V_f + (Z_{ext} + jX_{dg}^{'}) I_L$$ Regime permanente

$$I_g^{'} = \frac{E_g^{'}}{Z_{ext} + jX_{dg}^{'}} = \frac{V_f}{Z_{ext} + jX_{dg}^{'}} + I_L$$ Curto-circuito

Circuito para calcular *I"* com duas máquinas síncronas



$$E_g'' = V_f + (Z_{ext} + jX_{dg}'')I_L$$

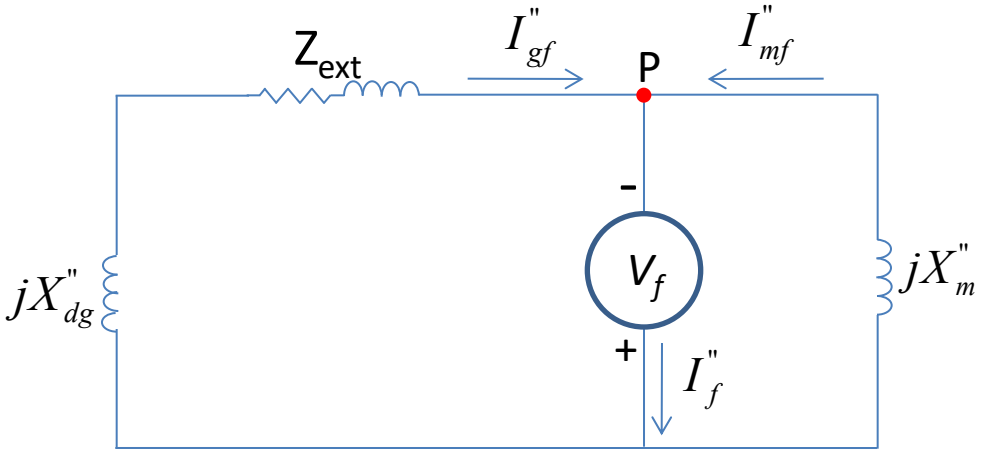
$$E_m'' = V_f - jX_{dm}''I_L$$

$$I_g^{"} = \frac{E_g^{"}}{Z_{ext} + jX_{dg}^{"}} = \frac{V_f}{Z_{ext} + jX_{dg}^{"}} + I_L$$

$$I_m^{"} = \frac{E_m^{"}}{jX_{dm,}^{"}} = \frac{V_f}{jX_{dm}^{"}} - I_L$$

$$I_f^{"} = I_g^{"} + I_m^{"} = \frac{V_f}{Z_{ext} + jX_{dg}^{"}} + \frac{V_f}{jX_{dm}^{"}} = I_{gf}^{"} + I_{mf}^{"}$$

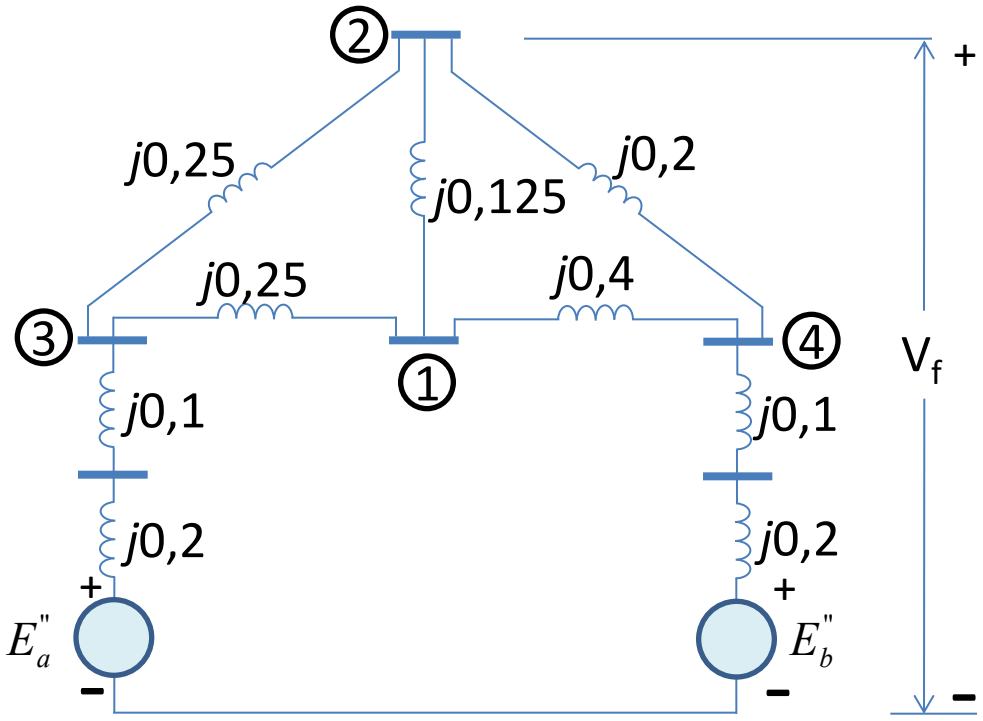
$$Z_{th} = \frac{jX_{dm}^{"}(Z_{ext} + jX_{dg}^{"})}{Z_{ext} + j(X_{dg}^{"} + X_{dm}^{"})}$$



$$I''_f = \frac{V_f}{Z_{th}} = \frac{V_f[Z_{ext} + j(X''_{dg} + X''_{dm})]}{jX''_{dm}(Z_{ext} + jX''_{dg})}$$

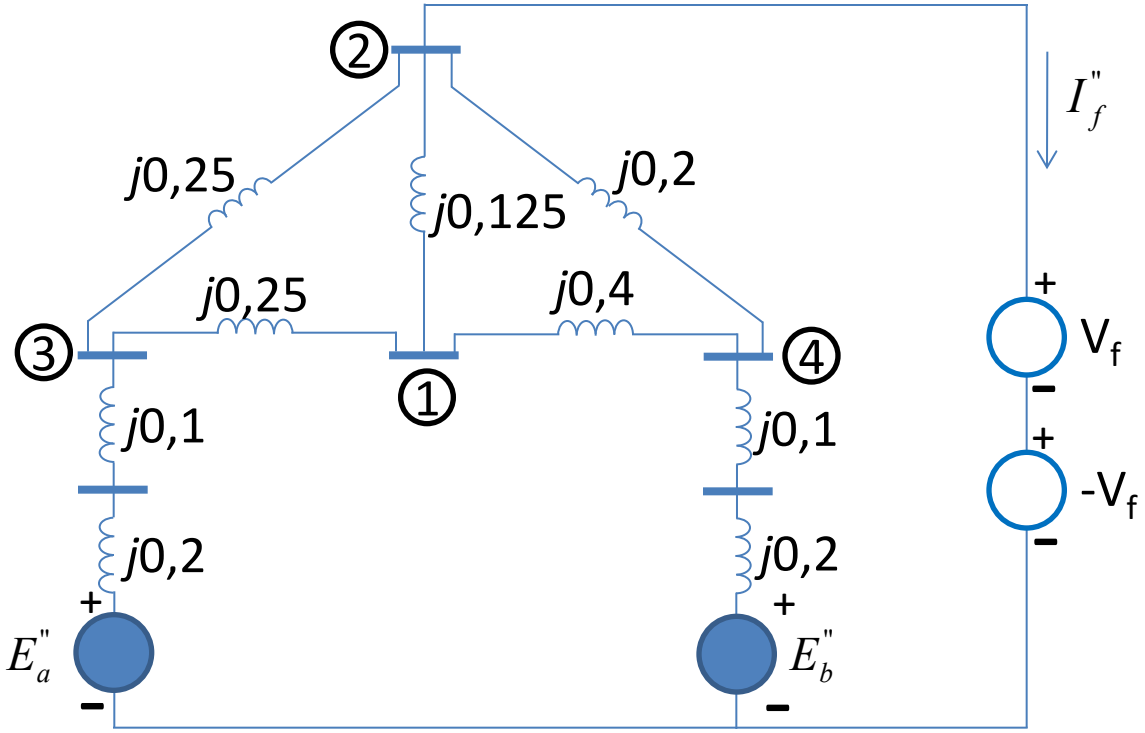
Dinâmica 6

Cálculo de faltas utilizando a Z_{BUS}



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Circuito com uma falta na barra 2



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$$\begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_3 \\ \Delta V_4 \end{bmatrix} = \begin{bmatrix} \Delta V_1 \\ -V_f \\ \Delta V_3 \\ \Delta V_4 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{14} \\ Z_{21} & Z_{22} & Z_{23} & Z_{24} \\ Z_{31} & Z_{32} & Z_{33} & Z_{34} \\ Z_{41} & Z_{42} & Z_{43} & Z_{44} \end{bmatrix} \begin{bmatrix} 0 \\ -I_f^{"} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -Z_{12} I_f^{"} \\ -Z_{22} I_f^{"} \\ -Z_{32} I_f^{"} \\ -Z_{42} I_f^{"} \end{bmatrix}$$

$$I_f^{"} = \frac{V_f}{Z_{22}}$$

$$\begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_3 \\ \Delta V_4 \end{bmatrix} = \begin{bmatrix} -\frac{Z_{12}}{Z_{22}} V_f \\ -V_f \\ -\frac{Z_{32}}{Z_{22}} V_f \\ -\frac{Z_{42}}{Z_{22}} V_f \end{bmatrix}$$

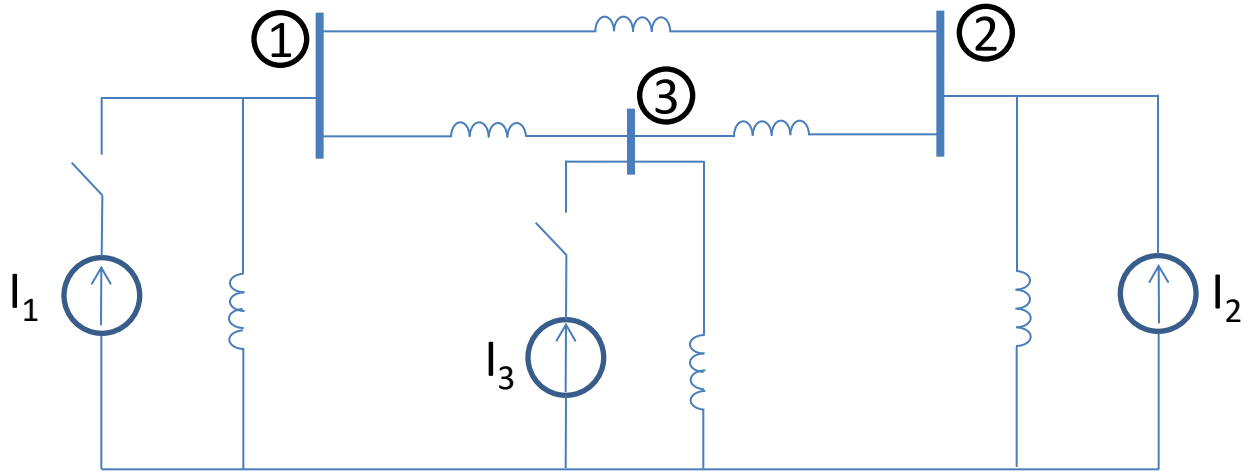
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$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} V_f \\ V_f \\ V_f \\ V_f \end{bmatrix} + \begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_3 \\ \Delta V_4 \end{bmatrix} = \begin{bmatrix} V_f - Z_{12} I_f^{"} \\ V_f - V_f \\ V_f - Z_{32} I_f^{"} \\ V_f - Z_{42} I_f^{"} \end{bmatrix} = V_f \begin{bmatrix} 1 - \dfrac{Z_{12}}{Z_{22}} \\ 0 \\ 1 - \dfrac{Z_{32}}{Z_{22}} \\ 1 - \dfrac{Z_{42}}{Z_{22}} \end{bmatrix}$$

$$I_f^{"} = \frac{V_f}{Z_{kk}} \qquad V_j = V_f - Z_{jk} I_f^{"} = V_f - \frac{Z_{jk}}{Z_{kk}} V_f \qquad I_{ij}^{"} = \frac{V_i - V_j}{Z_b}$$

Dinâmica 7

A matriz Z_{BUS}



$$\mathbf{V} = \mathbf{Z_{BUS}}\mathbf{I}$$

$$V_1 = Z_{11}I_1 + Z_{12}I_2 + Z_{13}I_3$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 + Z_{23}I_3$$

$$V_3 = Z_{31}I_1 + Z_{32}I_2 + Z_{33}I_3$$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=I_3=0}$$

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=I_3=0}$$

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Teorema de Thévenin

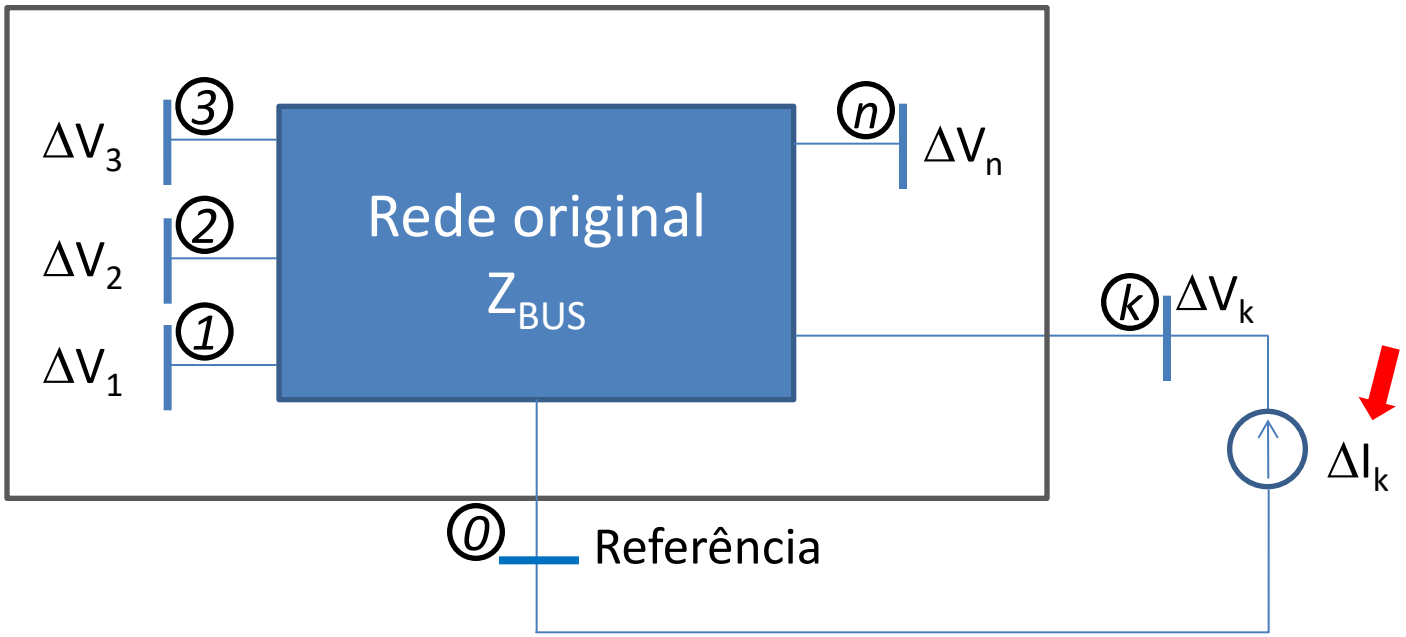
A impedância de Thévenin representa a rede em cada uma das barras.

Para uma corrente inicial $\mathbf{I}^0$:

$$\mathbf{V}^0 = \mathbf{Z}_{\mathbf{BUS}}\mathbf{I}^0$$

Se a corrente aumenta em $\Delta\mathbf{I}$:

$$\mathbf{V} = \mathbf{Z}_{\mathbf{BUS}}(\mathbf{I}^0 + \Delta\mathbf{I}) = \mathbf{Z}_{\mathbf{BUS}}\mathbf{I}^0 + \mathbf{Z}_{\mathbf{BUS}}\Delta\mathbf{I} = \mathbf{V}^0 + \Delta\mathbf{V}$$

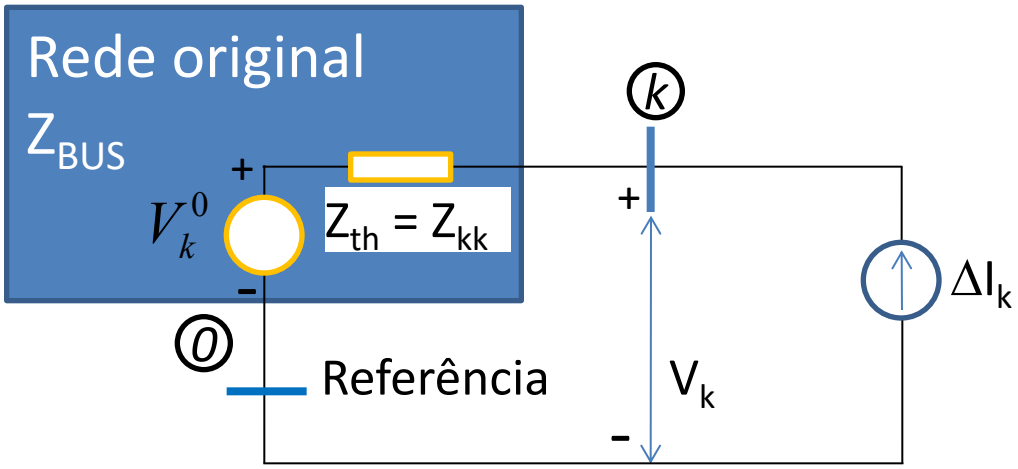

3
ΔV3
2
ΔV2
1
ΔV1
Rede original
ZBUS
n
ΔVn
k
ΔVk
ΔIk
0
Referência

$$\begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \vdots \\ \Delta V_k \\ \vdots \\ \Delta V_N \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1k} & \cdots & Z_{1N} \\ Z_{21} & Z_{22} & \cdots & Z_{2k} & \cdots & Z_{2N} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ Z_{k1} & Z_{k2} & \cdots & Z_{kk} & \cdots & Z_{kN} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ Z_{N1} & Z_{N2} & \cdots & Z_{Nk} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \Delta I_k \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} Z_{1k} \\ Z_{2k} \\ \vdots \\ Z_{kk} \\ \vdots \\ Z_{Nk} \end{bmatrix} \Delta I_k$$

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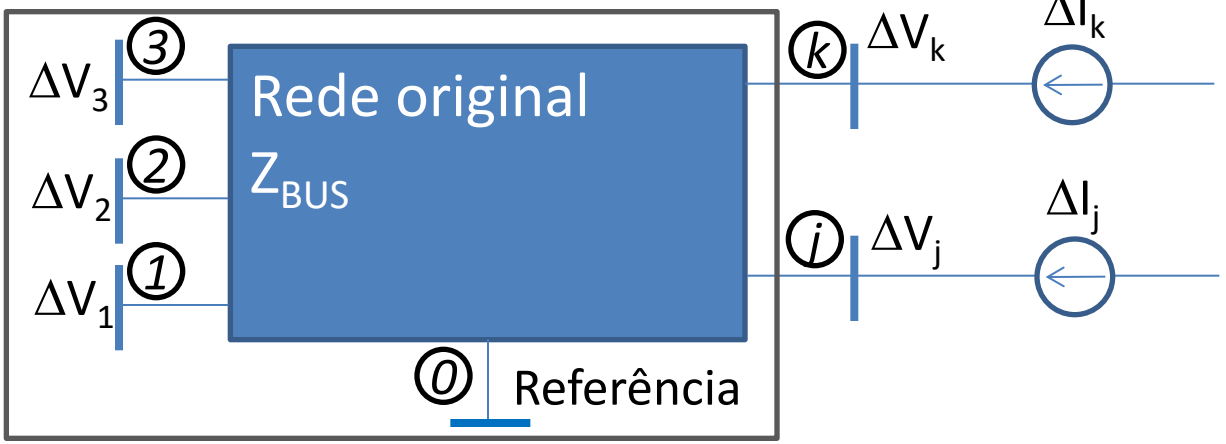
$$V_k = V_k^0 + \Delta V_k = V_k^0 + Z_{kk}\Delta I_k$$

$$Z_{th} = Z_{kk}$$



$$\begin{bmatrix} \Delta V_1 \\ \vdots \\ \Delta V_J \\ \Delta V_k \\ \vdots \\ \Delta V_N \end{bmatrix} = \begin{bmatrix} Z_{11} & \cdots & Z_{1j} & Z_{1k} & \cdots & Z_{1N} \\ \vdots & \ddots & \cdots & \vdots & \cdots & \vdots \\ Z_{j1} & \cdots & Z_{jj} & Z_{jk} & \ddots & Z_{jN} \\ Z_{k1} & \cdots & Z_{kj} & Z_{kk} & \cdots & Z_{kN} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_{N1} & \cdots & Z_{Nj} & Z_{Nk} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ \Delta I_j \\ \Delta I_k \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} Z_{1j}\Delta I_j + Z_{1k}\Delta I_k \\ \vdots \\ Z_{jj}\Delta I_j + Z_{jk}\Delta I_k \\ Z_{kj}\Delta I_j + Z_{kk}\Delta I_k \\ \vdots \\ Z_{Nj}\Delta I_j + Z_{Nk}\Delta I_k \end{bmatrix}$$

$$V_j = V_j^0 + Z_{jj}\Delta I_j + Z_{jk}\Delta I_k \qquad V_k = V_k^0 + Z_{kj}\Delta I_j + Z_{kk}\Delta I_k$$

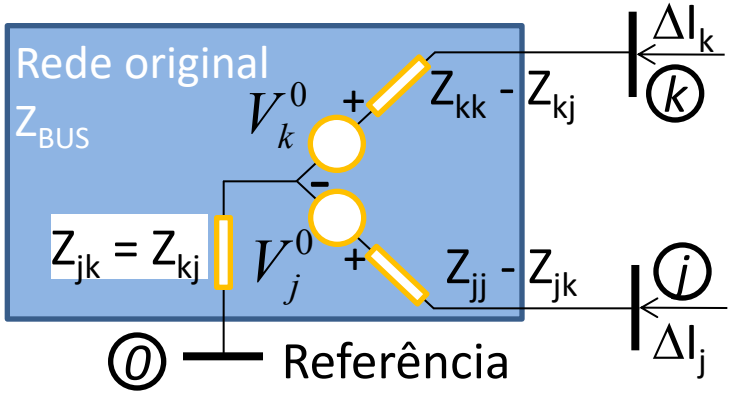


$$V_j = V_j^0 + Z_{jj}\Delta I_j + Z_{jk}\Delta I_k + Z_{jk}\Delta I_j - Z_{jk}\Delta I_j$$

$$V_k = V_k^0 + Z_{kj}\Delta I_j + Z_{kk}\Delta I_k + Z_{kj}\Delta I_k - Z_{kj}\Delta I_k$$

$$V_j = V_j^0 + (Z_{jj} - Z_{jk})\Delta I_j + Z_{jk}(\Delta I_j + \Delta I_k)$$

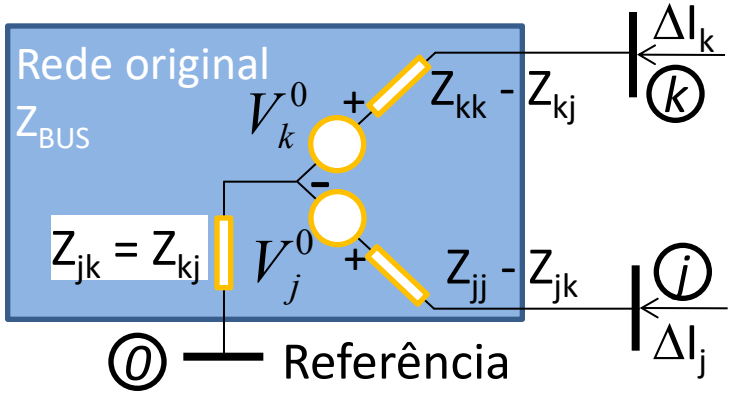
$$V_k = V_k^0 + Z_{kj}(\Delta I_j + \Delta I_k) + (Z_{kk} - Z_{kj})\Delta I_k$$



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$$V_j - V_k = [V_j^0 + (Z_{jj} - Z_{jk})\Delta I_j + Z_{jk}(\Delta I_j + \Delta I_k)] -$$
$$-[V_k^0 + Z_{kj}(\Delta I_j + \Delta I_k) + (Z_{kk} - Z_{kj})\Delta I_k]$$

$$V_j - V_k = V_j^0 - V_k^0 + (Z_{jj} - Z_{jk})\Delta I_j - (Z_{kk} - Z_{kj})\Delta I_k$$



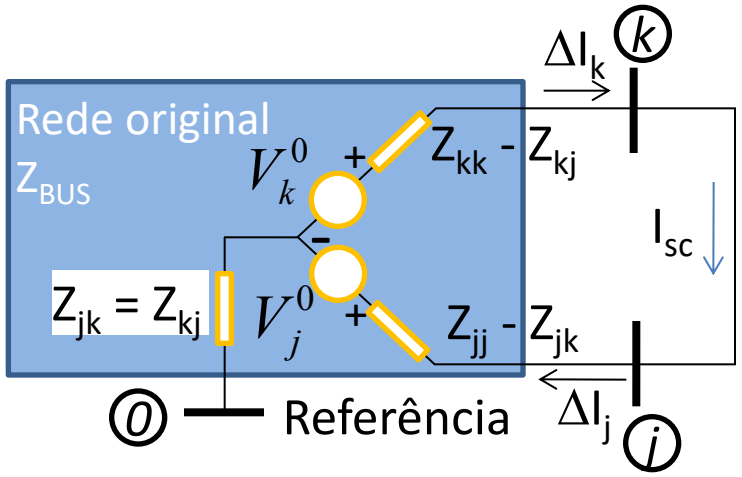
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Curto-circuitando as barras j e k:

$$0 = V_j^0 - V_k^0 + (Z_{jj} - Z_{jk})\Delta I_j - (Z_{kk} - Z_{kj})\Delta I_k \qquad \Delta I_j = -\Delta I_k = I_{sc}$$

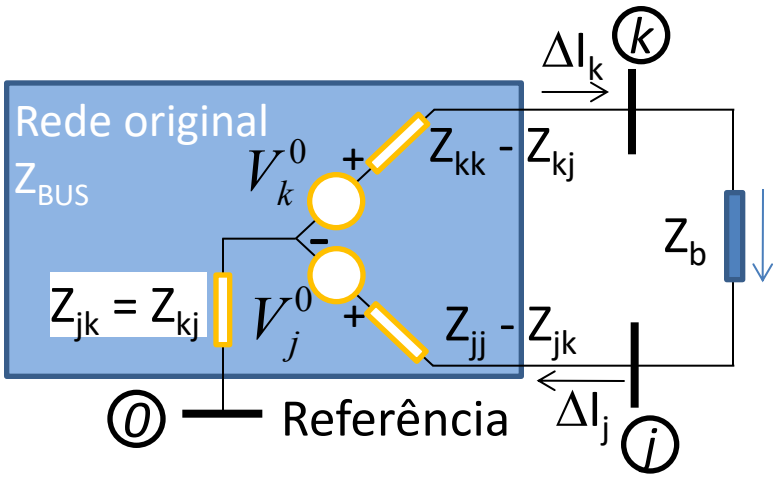
$$V_k^0 - V_j^0 = (Z_{jj} - Z_{jk} + Z_{kk} - Z_{kj})I_{sc} = Z_{th,jk} I_{sc}$$

$$Z_{th,jk} = Z_{jj} + Z_{kk} - 2Z_{jk}$$



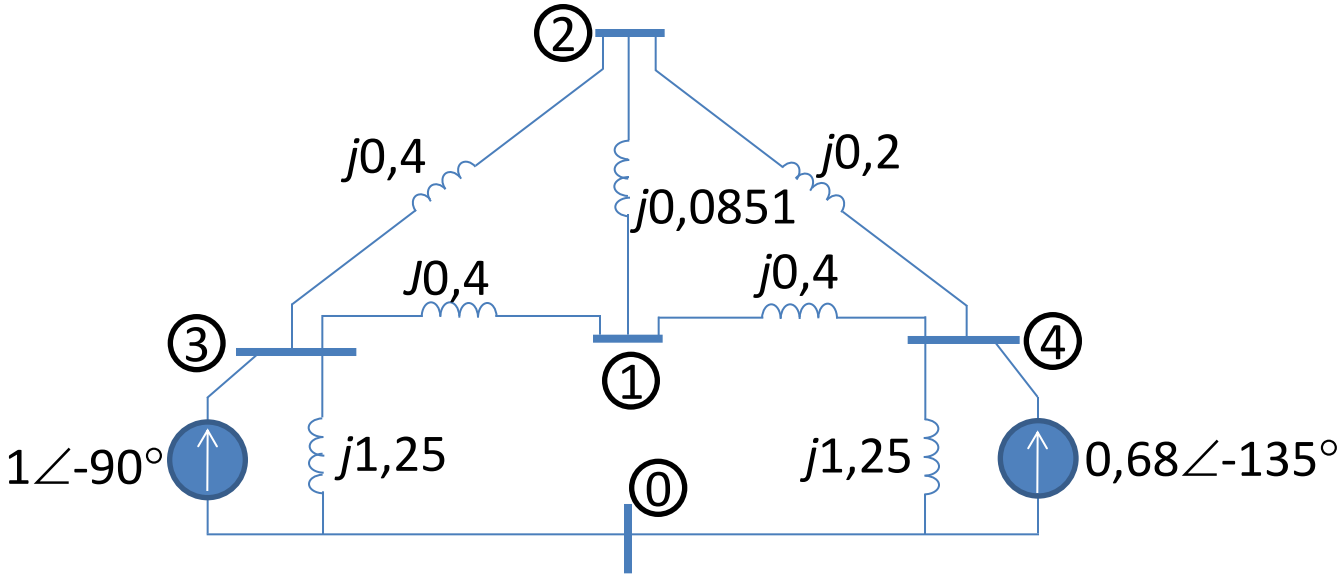
Conectando uma impedância entre as barras j e k:

$$V_k^0 - V_j^0 = \left(Z_{th,jk} + Z_b\right) I_b$$



$$I_b = \frac{V_k^0 - V_j^0}{Z_{th,jk} + Z_b} = \frac{V_k - V_j}{Z_b}$$

Exemplo: um capacitor que tem uma reatância de 5,0 pu é ligado entre a barra 4 e o terra. Encontre a corrente que passa pelo capacitor e a mudança de tensão em todas as barras.



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- Calcular as tensões iniciais nas barras:

$$\begin{bmatrix} V_1^0 \\ V_2^0 \\ V_3^0 \\ V_4^0 \end{bmatrix} = \begin{bmatrix} j0{,}73128 & j0{,}69140 & j0{,}61323 & j0{,}63677 \\ j0{,}69140 & j0{,}71966 & j0{,}60822 & j0{,}64178 \\ j0{,}61323 & j0{,}60822 & j0{,}69890 & j0{,}55110 \\ j0{,}63677 & j0{,}64178 & j0{,}55110 & j0{,}69890 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1\angle -90^o \\ 0{,}68\angle -135^o \end{bmatrix}$$

$$\begin{bmatrix} V_1^0 \\ V_2^0 \\ V_3^0 \\ V_4^0 \end{bmatrix} = \begin{bmatrix} 0{,}96903\angle -18{,}42^o \\ 0{,}96734\angle -18{,}6^o \\ 0{,}99964\angle -15{,}37^o \\ 0{,}94866\angle -20{,}75^o \end{bmatrix}$$

- A corrente que passa pelo capacitor é:

$$I_{cap} = \frac{V_4^0 - 0}{Z_{th,4} + Z_b} = \frac{0{,}94866\angle -20{,}75^\circ}{j0{,}69890 - j5} = 0{,}22056\angle 69{,}25^\circ$$

- Calcular a mudança na tensão provocada em todas as barras do sistema ao injetar a corrente anterior na barra 4

$$\begin{bmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_3 \\ \Delta V_4 \end{bmatrix} = \begin{bmatrix} j0{,}63677 \\ j0{,}64178 \\ j0{,}55110 \\ j0{,}69890 \end{bmatrix} (-0{,}22056\angle 69{,}25^\circ) = \begin{bmatrix} 0{,}14045\angle -20{,}75 \\ 0{,}14155\angle -20{,}75 \\ 0{,}12155\angle -20{,}75 \\ 0{,}15415\angle -20{,}75 \end{bmatrix}$$

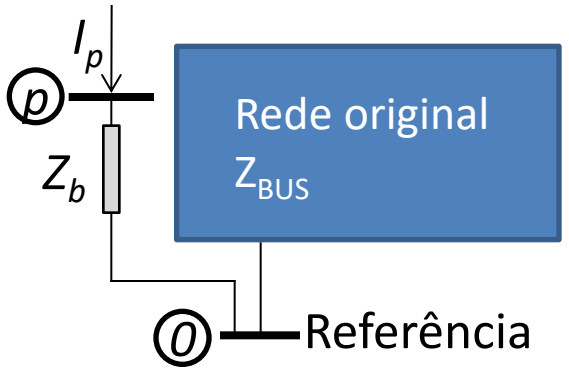
- A tensão resultante é:

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0{,}96903\angle -18{,}42^\circ \\ 0{,}96734\angle -18{,}6^\circ \\ 0{,}99964\angle -15{,}37^\circ \\ 0{,}94866\angle -20{,}75^\circ \end{bmatrix} + \begin{bmatrix} 0{,}14045\angle -20{,}75 \\ 0{,}14155\angle -20{,}75 \\ 0{,}12155\angle -20{,}75 \\ 0{,}15415\angle -20{,}75 \end{bmatrix} = \begin{bmatrix} 1{,}10938\angle -18{,}71^\circ \\ 1{,}10880\angle -18{,}88^\circ \\ 1{,}12071\angle -15{,}95^\circ \\ 1{,}10281\angle -20{,}75^\circ \end{bmatrix}$$

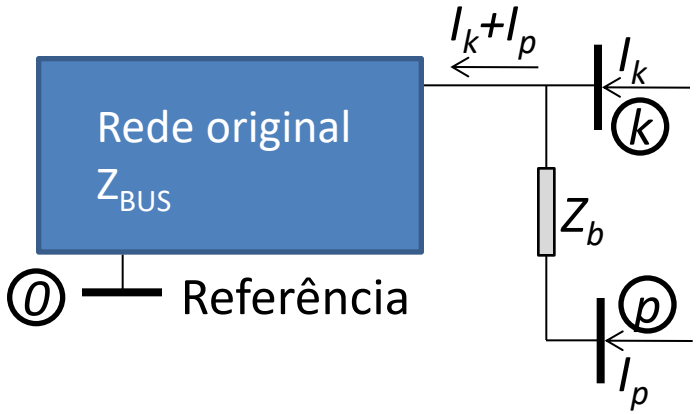
Modificação da matriz Z_{BUS}

1. Acrescentar a impedância Z_b de uma barra nova *p* ao nodo de referência

$$\begin{bmatrix} V_1^0 \\ V_2^0 \\ \vdots \\ V_N^0 \\ \hline V_p \end{bmatrix} = \left[\begin{array}{ccc|c} & Z_{orig} & & \begin{matrix}0\\0\\\vdots\\0\end{matrix} \\ \hline 0 & 0 \cdots & 0 & Z_b \end{array}\right] \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \\ \hline I_p \end{bmatrix}$$



2. Acrescentar a impedância Z_b de uma barra nova *p* a uma barra existente *k*



$$V_k = V_k^0 + I_p Z_{kk}$$ Slide 28

$$V_p = V_k^0 + I_p Z_{kk} + I_p Z_b$$

$$V_p = \underbrace{I_1 Z_{k1} + I_2 Z_{k2} + \cdots + I_N Z_{kN}}_{V_k^0} + I_p (Z_{kk} + Z_b)$$

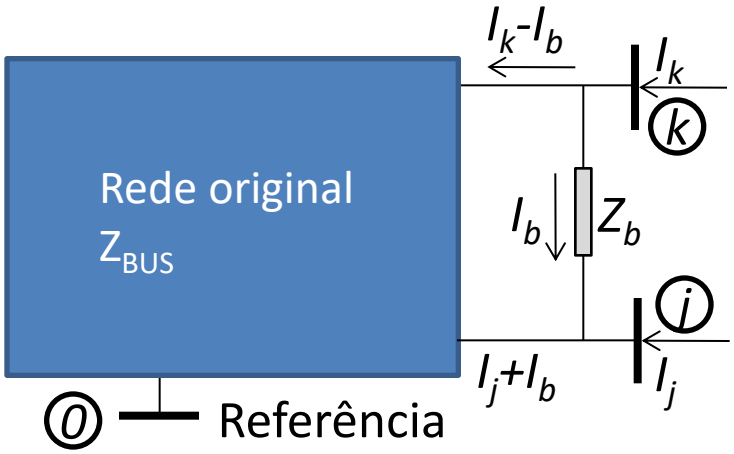
$$\begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \\ \hline V_p \end{bmatrix} = \left[\begin{array}{ccc|c} & Z_{orig} & & \begin{matrix} Z_{k1} \\ Z_{k2} \\ \vdots \\ Z_{kN} \end{matrix} \\ \hline Z_{k1} & Z_{k2} \quad \cdots \quad & Z_{kN} & Z_{kk} + Z_b \end{array}\right] \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \\ \hline I_p \end{bmatrix}$$

3. Acrescentar a impedância Z_b desde uma barra existente *k* ao nó de referência

- É acrescentada uma barra *p* conectada, através de Z_b, à barra *k*.
  - É obtida a mesma equação matricial do caso anterior.
- Se curto-circuita a barra *p* ao nó de referência fazendo V_p igual a zero.
- A fila e a coluna acrescentada é eliminada utilizando a redução de Kron.

$$Z_{hi(nova)} = Z_{hi} - \frac{Z_{h(N+1)} Z_{(N+1)i}}{Z_{kk} + Z_b}$$

4. Acrescentar a impedância Z_b entre duas barras existentes j e k



$$\begin{bmatrix} \Delta V_1 \\ \vdots \\ \Delta V_J \\ \Delta V_k \\ \vdots \\ \Delta V_N \end{bmatrix} = \begin{bmatrix} Z_{11} & \cdots & Z_{1j} & Z_{1k} & \cdots & Z_{1N} \\ \vdots & \ddots & \cdots & \vdots & \cdots & \vdots \\ Z_{j1} & \cdots & Z_{jj} & Z_{jk} & \ddots & Z_{jN} \\ Z_{k1} & \cdots & Z_{kj} & Z_{kk} & \cdots & Z_{kN} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_{N1} & \cdots & Z_{Nj} & Z_{Nk} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ \Delta I_j \\ \Delta I_k \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} Z_{1j}\Delta I_j + Z_{1k}\Delta I_k \\ \vdots \\ Z_{jj}\Delta I_j + Z_{jk}\Delta I_k \\ Z_{kj}\Delta I_j + Z_{kk}\Delta I_k \\ \vdots \\ Z_{Nj}\Delta I_j + Z_{Nk}\Delta I_k \end{bmatrix}$$

$$\Delta V_h = Z_{hj}\Delta I_j + Z_{hk}\Delta I_k$$

$$\Delta I_j = -\Delta I_k = I_b$$

$$\Delta V_h = (Z_{hj} - Z_{hk}) I_b$$

$$V_1 = V_1^0 + \Delta V_1$$

$$V_1 = Z_{11} I_1 + \cdots + Z_{1j} I_j + Z_{1k} I_k + \cdots + Z_{1N} I_N + (Z_{1j} - Z_{1k}) I_b$$

$$V_j = Z_{j1} I_1 + \cdots + Z_{jj} I_j + Z_{jk} I_k + \cdots + Z_{jN} I_N + (Z_{jj} - Z_{jk}) I_b$$

$$V_k = Z_{k1} I_1 + \cdots + Z_{kj} I_j + Z_{kk} I_k + \cdots + Z_{kN} I_N + (Z_{kj} - Z_{kk}) I_b$$

$$I_b = \frac{V_k^0 - V_j^0}{Z_{th,jk} + Z_b} \qquad 0 = V_j^0 - V_k^0 + (Z_{th,jk} + Z_b) I_b$$

$$V_j^0 = Z_{j1} I_1 + \cdots + Z_{jj} I_j + Z_{jk} I_k + \cdots + Z_{jN} I_N$$

$$V_k^0 = Z_{k1} I_1 + \cdots + Z_{kj} I_j + Z_{kk} I_k + \cdots + Z_{kN} I_N$$

$$0 = [(\text{fila } j \text{ - fila } k) \text{ de } Z_{\text{orig}}] \begin{bmatrix} I_1 \\ \vdots \\ I_j \\ I_k \\ \vdots \\ I_N \end{bmatrix} + (Z_{th,jk} + Z_b) I_b$$

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$$\begin{bmatrix} V_1 \\ \vdots \\ V_j \\ V_k \\ \vdots \\ V_N \\ \hline 0 \end{bmatrix} = \left[\begin{array}{c|c} Z_{\text{orig}} & \begin{array}{l}(\text{col. } j - \text{col. } k)\\ \text{de } Z_{\text{orig}}\end{array} \\ \hline \begin{array}{l}(\text{linha } j - \text{linha } k)\\ \text{de } Z_{\text{orig}}\end{array} & Z_{bb} \end{array}\right] \begin{bmatrix} I_1 \\ \vdots \\ I_j \\ I_k \\ \vdots \\ I_N \\ \hline I_b \end{bmatrix}$$

$$Z_{bb} = Z_{th,jk} + Z_b = Z_{jj} + Z_{kk} - 2Z_{jk} + Z_b$$

Dinâmica 8